

4.3 The impacts of tectonic hazards

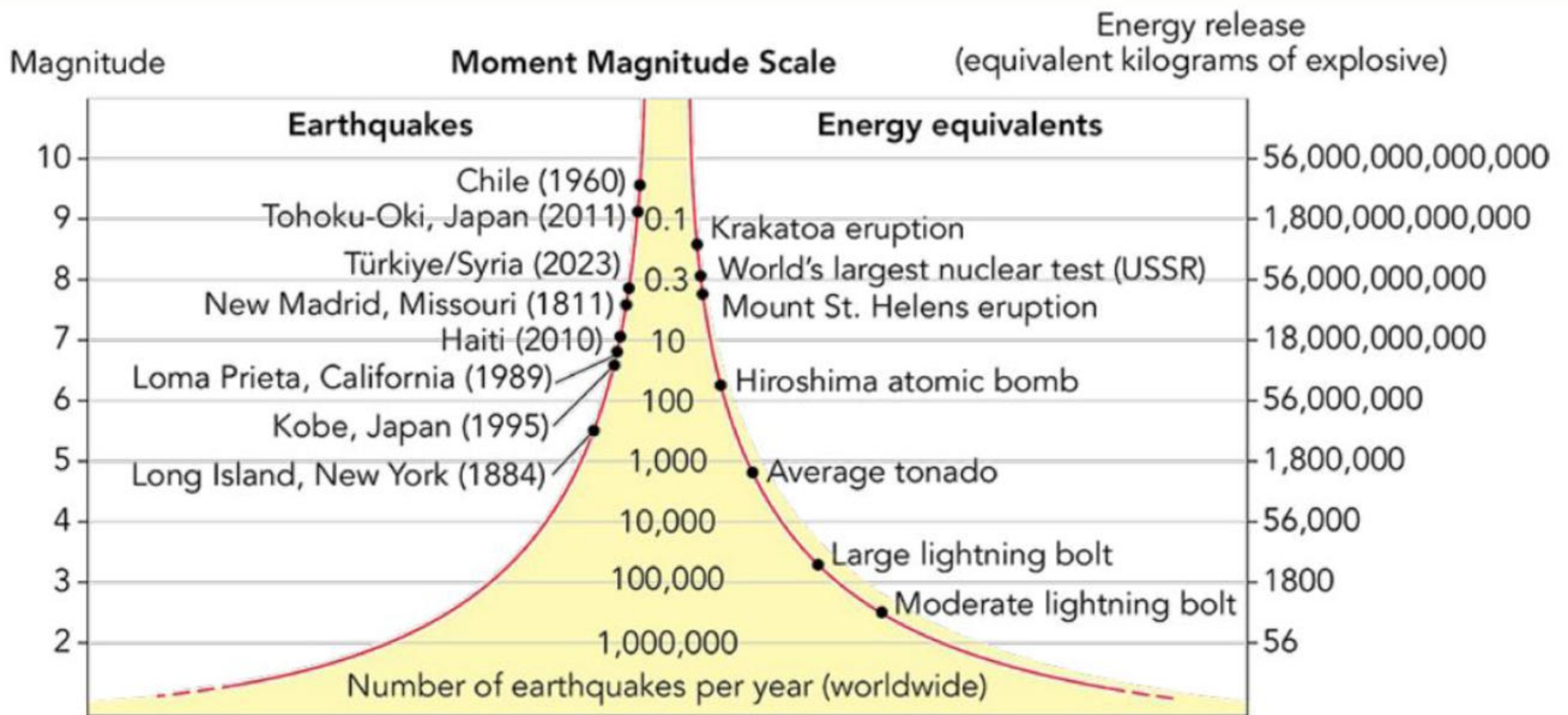
Tectonic hazards occur at both different magnitudes (sizes) and speeds. There are scales to measure how severe a tectonic hazard is. Scientists use a combination of different scales, depending on what they are trying to detect. The measurement tools in Table 4.2 allow scientists to measure the size of an earthquake or a volcanic eruption. Scientists can use this information to predict future tectonic events, and to compare the scale of the event and the damage caused in different locations.

Table 4.2: Measuring tools for earthquakes and volcanic eruptions

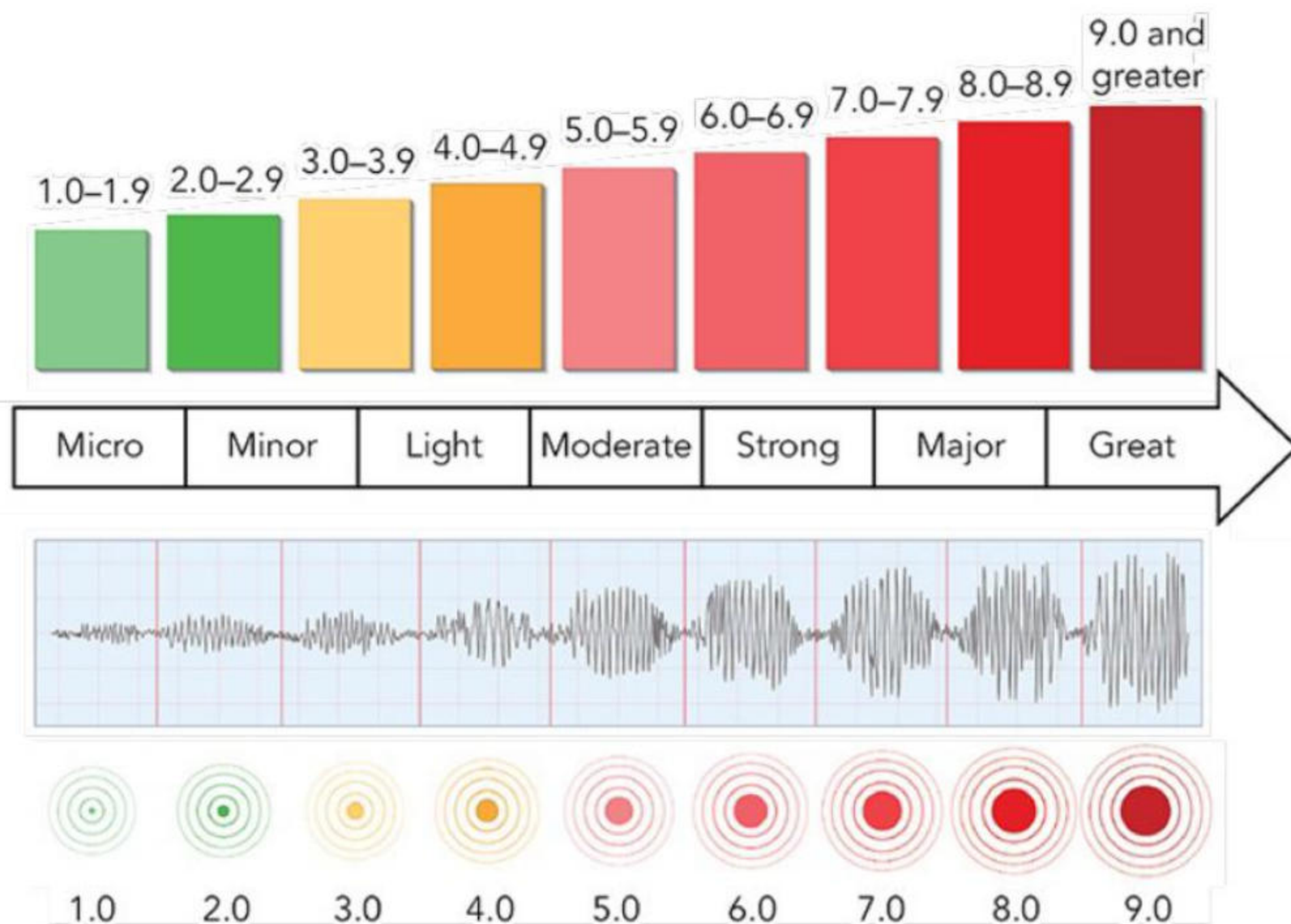
Measuring Earthquakes	
Measurement tool	Description
Moment Magnitude Scale	Modern seismometers and computer models allow for a far better analysis of the processes occurring during an earthquake. This resulted in the development of the Moment Magnitude Scale (MMS – Source A). What does the scale measure? The MMS measures the energy released and the distance the Earth shifts during an earthquake. The MMS defines how much force is needed to generate the recorded seismic waves. Where does the scale start and end? The MMS ranges from 1 at the smallest to 10 at the largest. How does the scale increase? This scale is logarithmic, meaning that a magnitude 6 earthquake has ten times more ground shaking than a magnitude 5 earthquake. Unique features This scale was developed in the 1970s and is more effective for measuring larger earthquakes than the Richter scale.
Richter scale	What does the scale measure? The Richter scale (Source B) is a measure of the magnitude of an earthquake, recording the amplitude of seismic waves. Where does the scale start and end? The Richter scale ranges from 1 at the smallest to 12 at the largest. How does the scale increase? Like the MMS, this scale is logarithmic. Unique features This scale was developed in the 1930s, and was the original scale developed using information gathered from seismographs.
Modified Mercalli intensity scale	What does the scale measure? The Modified Mercalli intensity scale (MMI–Source C) is based on the effects as they are reported by untrained observers of the earthquake. It does not require instruments. It reflects increasing levels of intensity that range from minimal shaking to complete destruction. Where does the scale start and end? The scale uses Roman numerals. The MMI ranges from I (1) at the smallest (‘Not felt’) to XII (12) at the largest (‘Extreme’). How does the scale increase? Unlike the MMS and the Richter, the MMI is not logarithmic. This means that its measures increase by a comparable amount (for example, the difference between 1 and 2 is comparable to the difference between 4 and 5). Unique features The Modified Mercalli intensity scale is the oldest of the three scales.
Measuring volcanic eruptions	
Volcanic Explosivity Index	What does the scale measure? The Volcanic Explosivity Index (VEI) describes the size of a volcanic eruption based on both magnitude and intensity. It uses an open-ended logarithmic scale. This means that the scale does not have a maximum measure. The VEI includes measurement of the volume of volcanic products, height of eruption cloud, and qualitative observations. Where does the scale start and end? A value of 0 is given for a non-explosive eruption and 8 represents a mega-colossal explosive eruption with 1000 km³ of volcanic cloud column of over 20 km. The scale does not have a maximum measure, but 8 represents the largest eruption in history. How does the scale increase? Like the MMS and Richter scale, the VEI is logarithmic. Unique features It is used to measure the explosiveness of a volcano, unlike the other tools that measure earthquake activity.

TIP

When trying to understand what logarithmic means, imagine the scales are like a volume knob, measuring how 'loud' an earthquake is. Each time the knob is turned up, it represents a change in 'loudness'. Different scales increase in 'loudness' by different amounts. For logarithmic scales, like the MMS and the Richter, the increases are by a factor of 10 respectively. Instead of a volume knob being turned an equal number of degrees, imagine the whole knob growing.



Source **A**: The Moment Magnitude Scale



Source **B**: The Richter scale

► **Image Description**

LEVEL	DESCRIPTION
I	Not felt except by a very few under especially favourable circumstances.

II	Felt only by a few persons at rest, especially on upper floors of buildings. Delicately suspended objects may swing.
III	Felt quite noticeably indoors, especially on upper of buildings, but many people do not recognise it as an earthquake. Standing motor cars may rock slightly. Vibration like passing of truck. Duration estimated.
IV	During the day felt indoors by many, outdoors by few. At night some awakened. Dishes, windows, doors disturbed; walls make cracking sound. Sensation like heavy truck striking building. Standing motor cars rocked noticeably.
V	Felt by nearly everyone, many awakened. Some dishes, windows, etc. broken; a few instances of cracked plaster, unstable objects overturned. Disturbances of trees, poles, and other tall objects sometimes noticed. Pendulum clocks may stop.
VI	Felt by all, many frightened and run indoors. Some heavy furniture moved; a few instances of fallen plaster or damaged chimneys. Damage slight.
VII	Everybody runs outdoors. Damage negligible in buildings of good design and construction; slight to moderate in well-built ordinary structures; considerable in poorly built or badly designed structures; some chimneys broken. Noticed by persons driving motor cars.
VIII	Damage slight in specially designed structures; considerable in ordinary substantial buildings, with partial collapse; great in poorly built structures. Panel walls thrown out of frame structures. Fall of chimneys, factory stacks, columns, monuments, walls. Heavy furniture overturned. Sand and mud ejected in small amounts. Changes in well water. Persons driving motor cars disturbed.
IX	Damage considerable in specially designed structures; well-designed frame structures thrown out of plumb; great in substantial building, with partial collapse. Building shifted off foundations. Ground cracked conspicuously. Underground pipes broken.
X	Some well-built wooden structures destroyed most masonry and frame structures destroyed with foundations; ground badly cracked. Rail bent. Landslides considerable from riverbanks and steep slopes. Shifted sand and mud. Water splashed (slopped) over banks.
XI	Few, if any, (masonry) structures remain standing. Bridges destroyed. Broad fissures in ground. Underground pipelines completely out of service. Earth slumps and land slips in soft ground. Rails bent greatly.
XII	Damage total. Practically all works of construction are damaged greatly or destroyed. Waves seen of ground surface. Lines of sight and level are distorted. Objects are thrown into the air.

Source : The Modified Mercalli intensity scale

TASK 1: Study Sources   

a

At what strength on the Modified Mercalli intensity scale is the earthquake felt by all, and may result in the movement of some heavy furniture or cause plaster to fall off the wall?

b

With reference to Source A, **describe the relationship** of the number of earthquakes per year (worldwide) with the magnitude of the earthquake.

c

Compare the Modified Mercalli intensity scale with the Moment Magnitude Scale and Richter scale.

Primary and secondary effects of earthquakes and volcanic eruptions

Natural disasters have both primary and secondary effects. The primary effects are the effects caused directly by either the Earth shaking or the volcanic eruption. Primary effects include the loss of life or number of people injured, for example. Secondary effects are the effects caused by the primary effects. They occur in the hours, days, weeks, or even years after the event and can include the spread of disease, the cost of rebuilding or out-migration of the population, and economic collapse in a region.

Both earthquakes and volcanic eruptions can cause death, injury, damage to property and infrastructure, damage to agriculture and food production, and can cause economic hardship. Several factors play a role in how significant the impacts are.

These are the:

- Strength of the earthquake or eruption:** the stronger and larger the event, the greater the potential impacts.
- Distance from human settlements:** the further the epicentre of an earthquake or the volcanic eruption is from human settlements, the lower the potential impact on humans.
- Time of day of an event:** when an earthquake occurs while people are sleeping, there is a greater risk of death or injury. People are slower to respond and more people are indoors, so collapsing buildings are more likely to cause harm.
- Frequency of events:** when a volcano erupts, or earthquakes occur regularly, the population is more likely to be prepared for such an event. When they occur hundreds or even thousands of years apart, the population is less likely to be prepared, so the impacts may be greater.
- Speed of the event:** the faster the event happens, the less time there is for people to get to safety. This increases the risk of injuries and death.

The impacts of volcanic eruptions are generally considered less severe than those caused by earthquakes. This is because volcanic eruptions are usually more predictable. Observation and monitoring of a volcano can give early warning that it is likely to erupt. Earthquakes are less predictable, however, which results in a higher risk of harm.

We can categorise the impacts of hazards as environmental, social, and economic. The environmental impacts are those that harm or benefit nature. For example, damaging plant life and destroying habitat. The social impacts are those that affect people and communities. For example, injuries or the loss of homes.

The economic impacts are those that have a cost linked to them. For example, the loss of a business and the income from it, or damage to a home and the cost of rebuilding.

The interconnectedness of these impacts is very important to consider. Economic, social, and environmental impacts all overlap each other. For example, volcanic soils are high in minerals and provide fertile soils, which are necessary for plant growth. This supports healthy habitat formation and higher biodiversity, which is a positive environmental impact. Fertile soil also allows for good growth of crops for farmers, allowing them to feed their families and support a healthy population. This is a positive social impact. Those farmers that produce more food than they need can sell their produce in the market. This is a positive economic impact. When considering impacts, one needs to think about how they may impact these three categories.

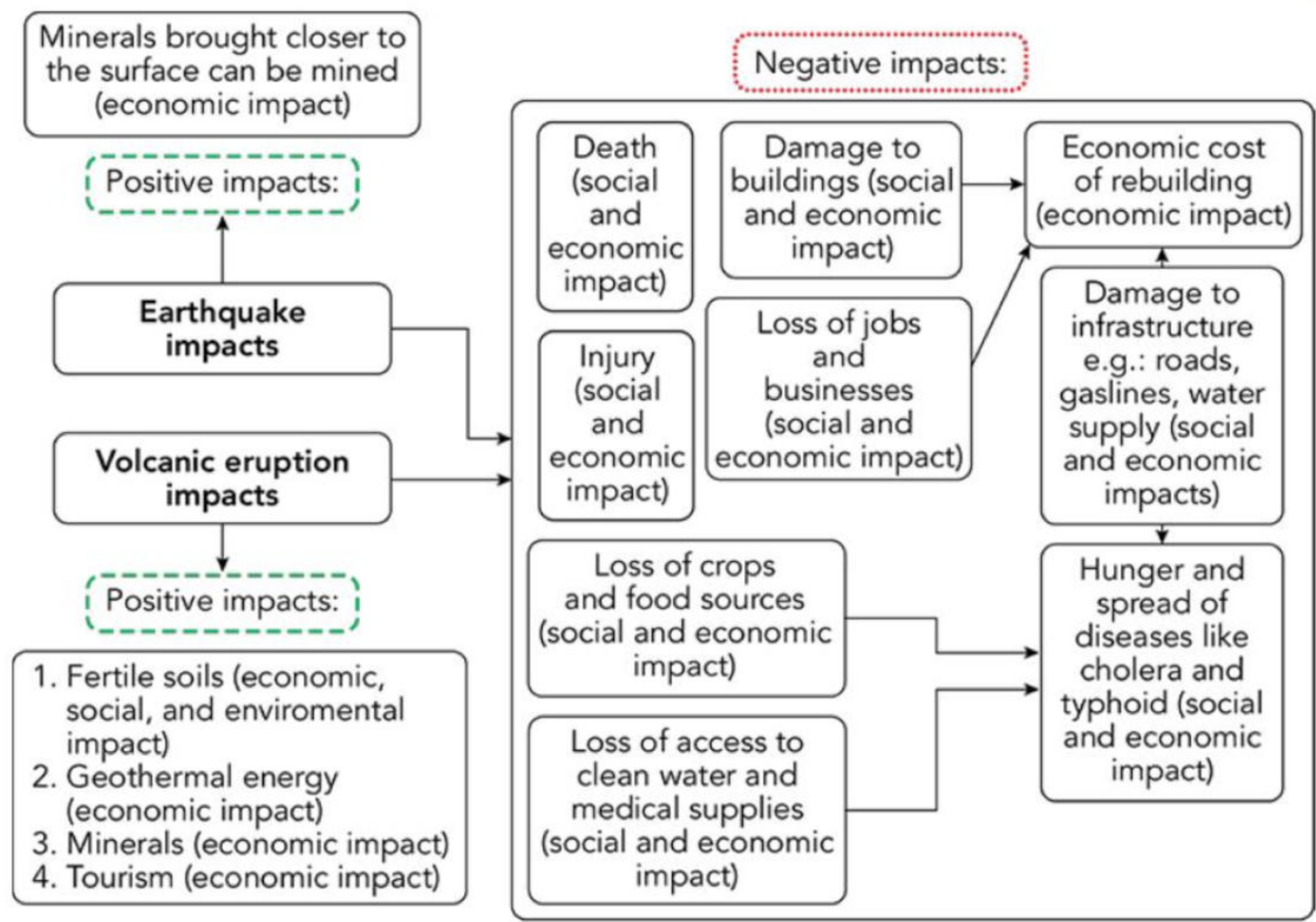


Figure 4.2: The impact of volcanoes and earthquakes

► Image Description

Volcanoes can trigger earthquakes and therefore share all the hazards that come with earthquakes. However, volcanic eruptions also have impacts that are specific to volcanic activity (Table 4.3).

Table 4.3: Volcano-specific impacts

Volcanic hazard	Negative impacts	Positive impacts
Ash fall	- causes respiratory difficulties - coats plants and reduces photosynthesis - builds up in heavy layers on roofs, causing them to collapse - poses a risk to aircraft as it damages their engines.	high in minerals that plants require for growth, it therefore makes volcanic soils fertile and good for growing crops.
Magma and lava	the flow of lava burns items in its path, causing damage and destruction.	the heat near the Earth surface offers the opportunity for geothermal energy to create electricity for the population living near the volcano.
Lahars and pyroclastic flows	the speed and distance lahars or pyroclastic flows can travel increases the risk of death or injury.	volcanic material is high in nutrients and adds to the soil fertility.
Toxic gases	causes respiratory distress and can result in death.	water vapour is one of the gases released during an eruption. It can benefit plant life as it adds water to the water cycle.

An additional benefit of volcanoes is economic. Volcanic areas are often very beautiful, attracting tourists to both visit the area and to observe an eruption. This allows the local population to develop a tourism industry that supports the economy.

TIP

Remember that impacts are not always negative. When considering them, think about potential positive impacts. The impacts of a volcanic eruption are a good example of this.

Despite not having the same benefits as volcanic eruptions, people live in areas of high earthquake activity. Reasons for this include the beauty of the area. It may be near an area that offers many resources such as water, food, mineral resources, and even transport routes. In addition, the time between events may be long, so people decide that the advantages outweigh the risks.

TASK 2

- Explain how the speed and frequency of tectonic hazards can result in an increase or decrease in potential impacts.
- Consider the positive and negative impacts of ash fall from a volcano. List a positive and a negative impact for the three categories: 'economic', 'social', and 'environmental'.
- In pairs, discuss why two earthquakes of the same size in different locations can cause death in one location, but not in the other.

During a visit to Mount Agung, Indonesian photographer Putu Sayoga said: 'I saw people live so close to the volcano. I was afraid at the time, but the people I met really did not seem to care'. Locals laughed and said, 'Oh yeah, there is a fire over there'. Putu suggests that because the villagers live so close to the volcano, they have a different understanding of it. The Indonesian government has installed early warning systems to evacuate the population if necessary. When asked about the possibility of evacuation, Putu describes locals saying 'that they do not want to live in other villages, because they still have livestock and farms in the villages that they come from'.

Source D: Living in the shadow of a volcano

TASK 3: Study Source D

- Why do you think the people who live on the slopes of the volcano are less afraid than visitors?
- The Indonesian government has put steps in place to evacuate an area if it is identified as becoming dangerous. Do you think that all the villagers are happy to leave their homes? Explain your answer.
- List two volcanic hazards that the villagers may face.
- Explain why living on or close to a volcano is beneficial to the farmers.

THINK LIKE A GEOGRAPHER

Geographers need to think of how a volcanic eruption can impact people far from the site of an event. How do you think a volcanic eruption in Iceland may impact people as far away as Japan, the USA, and Australia? Consider how ash moves in the atmosphere.